a Precision support and measured unit area

31 selectable-mode implementations · K = 16 · Target clock = 1410 MHz

1 mode | 2 modes | 3 modes | 4 modes | 5 modes | 6 modes | 7 modes | 9 modes

Unit area (10^3 µm²)

Combined FP (7 modes): 23.9

0.18, 0.18, 2.0, 4.0, 4.7, 4.9, 5.5, 5.8, 6.2, 6.5, 7.0, 7.3, 12.8, 14.5, 14.6, 5.3, 8.4, 8.6, 9.7, 10.0, 20.0, 20.5, 9.7, 12.3, 20.6, 20.7, 23.2, 23.6, 23.9, 28.6, 29.1

TF32
FP16
BF16
FP8 (E4M3)
FP8 (E5M2)
FP6 (E3M2)
FP6 (E2M3)
FP4 (E2M1)
INT8
INT4
Binary (1 bit)

MX scaling
NV scaling
FP4 LUT
Packed 2×FP4

U01 n4, U02 n8, U03, U04, U05, U06, U07, U08, U09, U10, U11, U12, U13, U14, U15, U16, U17, U18, U19, U20, U21, U22, U23, U24, U25, U26, U27, U28, U29, U30, U31

Measured implementation (one column = one synthesized unit)

b Asymmetric FP × INT · K = 16

Area (10^3 µm²)

7.1, 10.6, 5.9, 7.1

FP16 × INT4, FP16 × INT8, FP8 × INT4, FP8 × INT8

c Additional units · K = 32

Area (10^3 µm²)

19.4, 16.4

FP8 + FP6 + 2×FP4, MXFP8 + FP6 + FP4

d Outer-product arrays · 8×8, K = 1

Area (10^3 µm²)

95.7, 107.2, 112.9, 238.1

BF16, FP16, TF32, FP32

LUT implementation | 9-mode combined units | Numeric mode | Implementation feature